

Online Retail Sales Analysis

Excel & Power BI Project

1. Project Overview and Objective

This project takes a raw e-commerce transaction dataset for a UK based online gift retailer, cleans and transforms it in Excel, and builds an interactive Power BI dashboard to analyze sales performance, product popularity, and customer buying behaviour.

The main objective is to demonstrate data pre-processing techniques in Excel, including handling missing values, removing duplicates and invalid records, and standardizing inconsistent formats, and then to convert the cleaned data into a star-schema data model with pivot tables and DAX measures to enable accurate sales, product and customer analysis, supported by a three-page interactive Power BI report with click-through storytelling navigation to make informed business decisions.

2. Data Sources

Source Description and Timeline: Online Retail dataset, UCI Machine Learning Repository, 1 December 2010 to 9 December 2011.

Domain: Online Retail Sales Analytics – UK based retailer selling unique all-occasion gift items.

<https://uci-ics-mlr-prod.aws.uci.edu/dataset/352/online%2Bretail?utm>

Attribute	Detail
Rows before cleaning	541,909 rows, 8 columns
Rows after cleaning	392,692 rows, 12 columns
Grain	One row per product line item, per invoice
Time period	1 December 2010 – 9 December 2011

3. Problem Statement

1. Descriptive Analytics (What is happening?)

- Identify the highest revenue generating countries and top-selling products by total sales value.
- Track total sales, total orders, distinct customers and total quantity sold across the full time period.
- Rank countries by revenue to pinpoint the highest and lowest performing markets.

2. Diagnostic Analytics (Why is it happening?)

- Market Concentration Analysis: Examine why the United Kingdom accounts for the large majority of total sales compared to other countries served.
- Product Mix Diagnosis: Cross-reference top-selling products with quantity sold to understand whether high revenue is driven by high-value or high-volume items.

- Customer Value Diagnosis: Identify whether a small group of customers is responsible for a disproportionate share of total revenue.

3. Predictive Analytics (What is likely to happen?)

- Use the recurring monthly sales trend to project likely sales levels going into future periods.
- Identify which countries and products are likely to continue driving the majority of future revenue if current buying patterns persist.

4. Prescriptive Analytics (What should we do about it?)

- Which customers should be prioritized for retention, given that a small group generates a large share of total revenue?
- Which secondary markets (e.g. Netherlands, EIRE, Germany, France) should receive increased marketing focus to grow sales beyond the UK?

4. Tools & Technologies

- Excel: Data cleaning, data transformation, and pivot tables.
- Power BI Desktop: Data modelling, DAX measures, and interactive report building.
- Power BI Service: Publishing the report online and configuring scheduled refresh.
- UCI Machine Learning Repository: Source of the raw dataset.

5. Data Pre-Processing

Tasks Performed

- Data Cleaning & Transformation: Removed duplicates, removed cancelled orders and rows with missing values, standardized formats, and added calculated fields.

Get Data — Loading the Source File

The cleaned Excel (Online_Retail_Final) workbook was loaded into Power BI using Get Data, shown below with the source file selected before import.

Today			
 Online_Retail_Final	05-07-2026 20:25	Microsoft Excel Work...	29,582 KB
 Online_Retail_Cleaned	05-07-2026 20:24	Microsoft Excel Work...	22,967 KB

File name:
Excel Files (*.xl;*.xlsx;*.xlsm;*.xlt)

Open
Cancel

Get Data dialog – source file selection before import

Removing Duplicates:

The screenshot shows a WPS Spreadsheet window with a data table. A dialog box titled "Microsoft Excel" is displayed in the center, stating "No duplicate values found." with an "OK" button. The data table has columns: Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country, Sales, Year, Month Name, and Quarter. The data is sorted by Sales in descending order. The status bar at the bottom indicates "Count: 1388".

	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Sales	Year	Month Name	Quarter
2	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010	1.85	15311	United Kingdom	15.30	2010	December	Q4
3	WHITE METAL LANTERN	6	01-12-2010	1.69	16098	United Kingdom	20.34	2010	December	Q4
4	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010	2.10	18074	United Kingdom	22.00	2010	December	Q4
5	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010	2.10	17420	United Kingdom	20.34	2010	December	Q4
6	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010	3.75	16029	United Kingdom	20.34	2010	December	Q4
7	SET 7 BABUSHKA NESTING BOXES	2	01-12-2010	1.65	16250	United Kingdom	15.30	2010	December	Q4
8	GLASS STAR FROSTED T-LIGHT HOLDER	6	01-12-2010	4.25	12431	United Kingdom	25.50	2010	December	Q4
9	HAND WARMER UNION JACK	6	01-12-2010	4.95	17511	United Kingdom	11.10	2010	December	Q4
10	HAND WARMER RED POLKA DOT	6	01-12-2010	4.25	12431	United Kingdom	11.10	2010	December	Q4
11	ASSORTED COLOUR BIRD ORNAMENT	32	01-12-2010	1.69	16098	United Kingdom	54.08	2010	December	Q4
12	POPPY'S PLAYHOUSE BEDROOM	6	01-12-2010	2.10	18074	United Kingdom	12.60	2010	December	Q4
13	POPPY'S PLAYHOUSE KITCHEN	6	01-12-2010	2.10	17420	United Kingdom	12.60	2010	December	Q4
14	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	01-12-2010	3.75	16029	United Kingdom	30.00	2010	December	Q4
15	IVORY KNITTED MUG COSY	6	01-12-2010	1.65	16250	United Kingdom	9.90	2010	December	Q4
16	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	01-12-2010	4.25	12431	United Kingdom	25.50	2010	December	Q4
17	BOX OF VINTAGE JIGSAW BLOCKS	3	01-12-2010	4.95	17511	United Kingdom	14.85	2010	December	Q4
18	BOX OF VINTAGE ALPHABET BLOCKS	2	01-12-2010	9.95	13705	United Kingdom	19.90	2010	December	Q4
19	HOME BUILDING BLOCK WORD	3	01-12-2010	5.95	13747	United Kingdom	17.85	2010	December	Q4

Standardized Format

Below showing as example, as formatted Date column as Date data type.

The screenshot shows a WPS Spreadsheet window with a data table. The 'InvoiceDate' column is formatted as a Date. The data table has columns: Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country, Sales, Year, Month Name, and Quarter. The data is sorted by Sales in descending order. The status bar at the bottom indicates "Average: 16-09-2011" and "Count: 261293".

	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Sales	Year	Month Name	Quarter
2	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010	2.55	17850	United Kingdom	15.30	2010	December	Q4
3	WHITE METAL LANTERN	6	01-12-2010	3.39	13047	United Kingdom	20.34	2010	December	Q4
4	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010	2.75	12583	United Kingdom	22.00	2010	December	Q4
5	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010	3.39	13748	United Kingdom	20.34	2010	December	Q4
6	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010	3.39	15100	United Kingdom	20.34	2010	December	Q4
7	SET 7 BABUSHKA NESTING BOXES	2	01-12-2010	7.65	15291	United Kingdom	15.30	2010	December	Q4
8	GLASS STAR FROSTED T-LIGHT HOLDER	6	01-12-2010	4.25	14688	United Kingdom	25.50	2010	December	Q4
9	HAND WARMER UNION JACK	6	01-12-2010	1.85	17809	United Kingdom	11.10	2010	December	Q4
10	HAND WARMER RED POLKA DOT	6	01-12-2010	1.85	15311	United Kingdom	11.10	2010	December	Q4
11	ASSORTED COLOUR BIRD ORNAMENT	32	01-12-2010	1.69	16098	United Kingdom	54.08	2010	December	Q4
12	POPPY'S PLAYHOUSE BEDROOM	6	01-12-2010	2.10	18074	United Kingdom	12.60	2010	December	Q4
13	POPPY'S PLAYHOUSE KITCHEN	6	01-12-2010	2.10	17420	United Kingdom	12.60	2010	December	Q4
14	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	01-12-2010	3.75	16029	United Kingdom	30.00	2010	December	Q4
15	IVORY KNITTED MUG COSY	6	01-12-2010	1.65	16250	United Kingdom	9.90	2010	December	Q4
16	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	01-12-2010	4.25	12431	United Kingdom	25.50	2010	December	Q4
17	BOX OF VINTAGE JIGSAW BLOCKS	3	01-12-2010	4.95	17511	United Kingdom	14.85	2010	December	Q4
18	BOX OF VINTAGE ALPHABET BLOCKS	2	01-12-2010	9.95	13705	United Kingdom	19.90	2010	December	Q4
19	HOME BUILDING BLOCK WORD	3	01-12-2010	5.95	13747	United Kingdom	17.85	2010	December	Q4

Conditional Formatting:

Showing the conditional formatting, In Total price column top 10 values are highlighted with pink color and filtered to find highest value of order.

File Home WPS PDF Insert Draw Page Layout Formulas Data Review View Automate Help									
Clipboard Font Alignment Number Styles Cells Editing Sensitivity Add-ins WPS PDF be amazing									
H11 : fx United Kingdom									
	B	C	D	E	F	G	H	I	J
1	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Sales	Y
2	85123A	WHITE HANGING HEART T-LIGHT HOLDER	80995	01-12-2010	2.55	17850	United Kingdom	1,68,469.60	
3	71053	WHITE METAL LANTERN	74215	01-12-2010	3.39	13047	United Kingdom	77,183.60	
4	84406B	CREAM CUPID HEARTS COAT HANGER	4800	01-12-2010	2.75	12583	United Kingdom	38,970.00	
5	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	4300	01-12-2010	3.39	13748	United Kingdom	7,144.72	
6	84029E	RED WOOLLY HOTTIE WHITE HEART.	3186	01-12-2010	3.39	15100	United Kingdom	4,992.00	
7	22752	SET 7 BABUSHKA NESTING BOXES	3000	01-12-2010	7.65	15291	United Kingdom	4,401.00	
8	21730	GLASS STAR FROSTED T-LIGHT HOLDER	2880	01-12-2010	4.25	14688	United Kingdom	4,176.00	
9	22633	HAND WARMER UNION JACK	2700	01-12-2010	1.85	17809	United Kingdom	4,161.06	
10	22632	HAND WARMER RED POLKA DOT	2400	01-12-2010	1.85	15311	United Kingdom	4,161.06	
11	84879	ASSORTED COLOUR BIRD ORNAMENT	2100	01-12-2010	1.69	16098	United Kingdom	3,949.32	
12	22745	POPPY'S PLAYHOUSE BEDROOM	2040	01-12-2010	2.10	18074	United Kingdom	3,861.00	
13	22748	POPPY'S PLAYHOUSE KITCHEN	2000	01-12-2010	2.10	17420	United Kingdom	3,825.36	
14	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	1992	01-12-2010	3.75	16029	United Kingdom	3,651.60	
15	22310	IVORY KNITTED MUG COSY	1944	01-12-2010	1.65	16250	United Kingdom	3,213.60	
16	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	1944	01-12-2010	4.25	12431	United Kingdom	3,321.00	
17	22623	BOX OF VINTAGE JIGSAW BLOCKS	1900	01-12-2010	4.95	17511	United Kingdom	3,285.00	

Creating New Columns for Analysis Requirement

- Create a calculated Sales column from Quantity and Unit Price

=ROUND(Quantity*UnitPrice,2)

- Creating Year, Month Name and Quarter columns from InvoiceDate for reporting

Year = YEAR(InvoiceDate) Month Name = TEXT(InvoiceDate,"mmm") Quarter = "Q"&ROUNDUP(MONTH(InvoiceDate)/3,0)

The Sales, Year, Month Name and Quarter formulas were converted to fixed values once verified, and the final row count (392,692) was checked using =COUNTA(A:A)-1, with pivot tables used to spot-check totals before moving the data into Power BI

6. Data Modelling and DAX (Power BI)

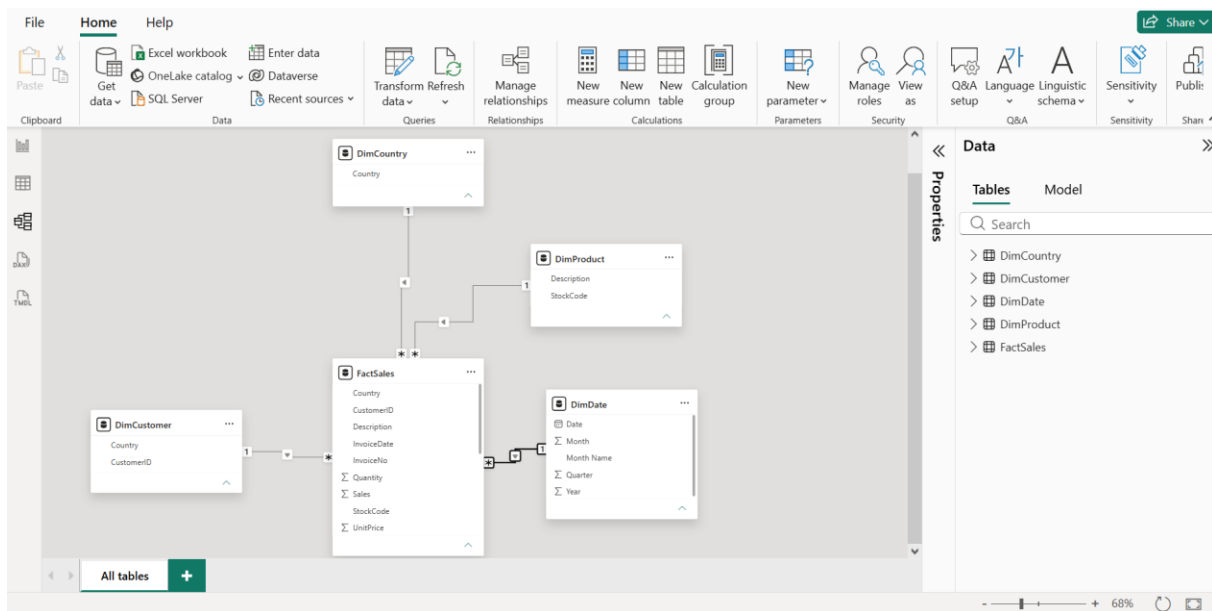
Manage relationships

+ New relationship Autodetect Edit Delete Filter				
☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ FactSales (Country)	1	DimCountry (Country)	Active	...
☐ FactSales (CustomerID)	1	DimCustomer (CustomerID)	Active	...
☐ FactSales (InvoiceDate)	1	DimDate (Date)	Active	...
☐ FactSales (StockCode)	1	DimProduct (StockCode)	Active	...

Close

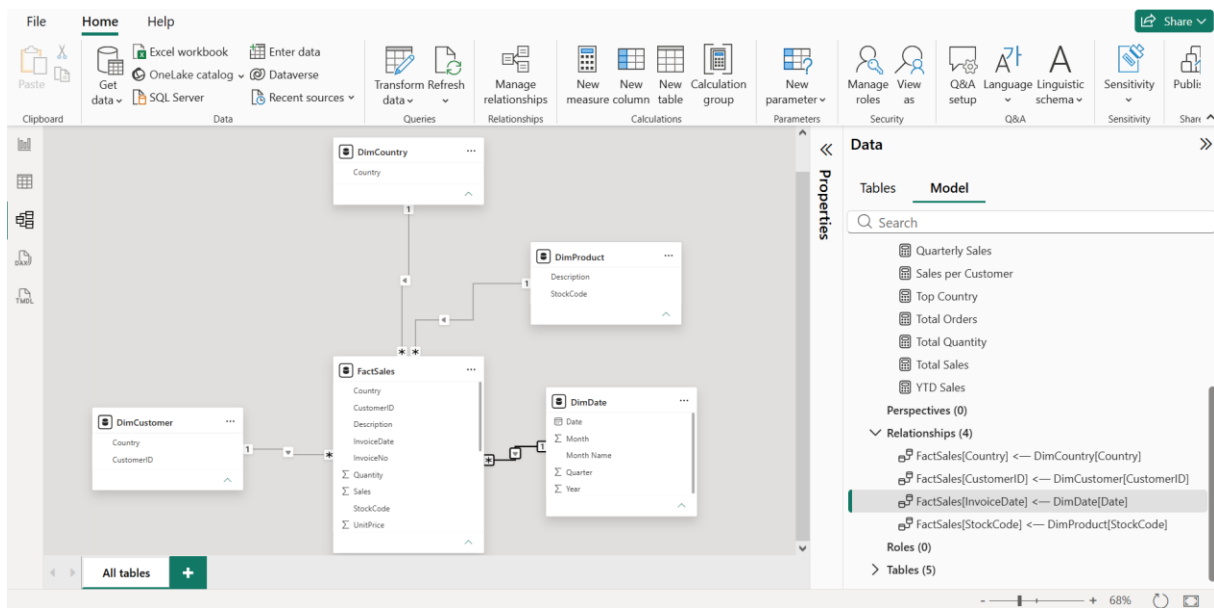
Data Model: A star schema was built with one fact table (FactSales) and four dimension tables (DimCustomer, DimProduct, DimDate, DimCountry), joined on active one-to-many relationships.

Fact Table Column	Dimension Table	Related Column
FactSales (Country)	DimCountry	Country
FactSales (CustomerID)	DimCustomer	CustomerID
FactSales (InvoiceDate)	DimDate	Date
FactSales (StockCode)	DimProduct	StockCode



Model view – FactSales connected to all four dimension tables

- FactSales into DimProduct: connected by StockCode, to view product-wise revenue and identify top-selling items. Type: Many-to-One.
- FactSales into DimCountry: connected by Country, to view country-wise sales performance and market concentration. Type: Many-to-One.
- FactSales into DimCustomer: connected by CustomerID, to analyze distinct customers and sales per customer. Type: Many-to-One.
- FactSales into DimDate: connected by InvoiceDate/Date, to analyze monthly, quarterly and yearly sales trends and drill down from year to month. Type: Many-to-One.

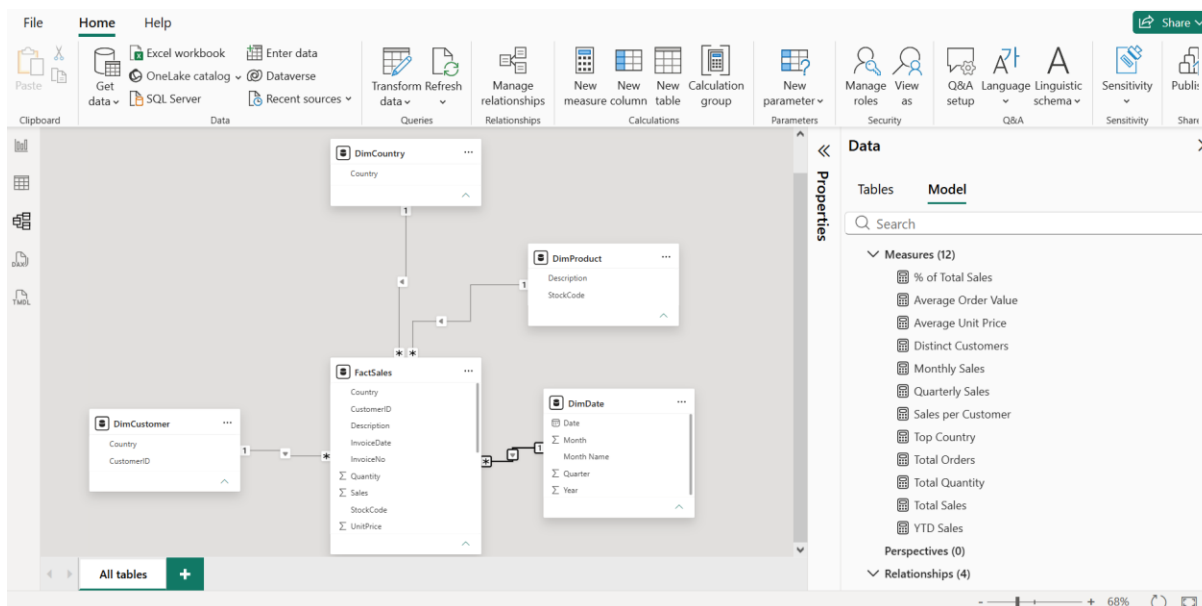


Manage Relationships – all four relationships shown as Active

Calculated Columns & DAX Measures

Twelve DAX measures were created for the report.

DAX Measure	Used
Total Sales	Overall revenue KPI for the period
Total Orders	Distinct count of invoices, to track order volume
Total Quantity	Total units sold, for demand analysis
Distinct Customers	Number of unique customers, for reach analysis
Average Order Value	Total Sales divided by Total Orders
Average Unit Price	Average price per unit sold
Sales per Customer	Total Sales divided by Distinct Customers
Monthly / Quarterly / YTD Sales	Time intelligence measures for trend analysis
% of Total Sales	Share of revenue by country or product
Top Country	Highest revenue-generating country
YTD Sales	Time intelligence measure - cumulative sales from start of year to current date
Sales per Customer	Total Sales divided by Distinct Customers



Data pane – all 12 measures listed under FactSales

Dax Measure	Attribute	Used
Total Sales	SUM(FactSales[Sales])	Overall revenue KPI for the period
Total Orders	DISTINCTCOUNT(FactSales[InvoiceNo])	Distinct count of invoices, to track order volume
Distinct Customers	DISTINCTCOUNT(FactSales[CustomerID])	Number of unique customers, for reach analysis
Average Order Value	DIVIDE([Total Sales], [Total Orders], 0)	Total Sales divided by Total Orders
% of Total Sales	DIVIDE(SUM(FactSales[Sales]), CALCULATE(SUM(FactSales[Sales]), ALL(FactSales)), 0)	Share of revenue by country or product

Formula Used

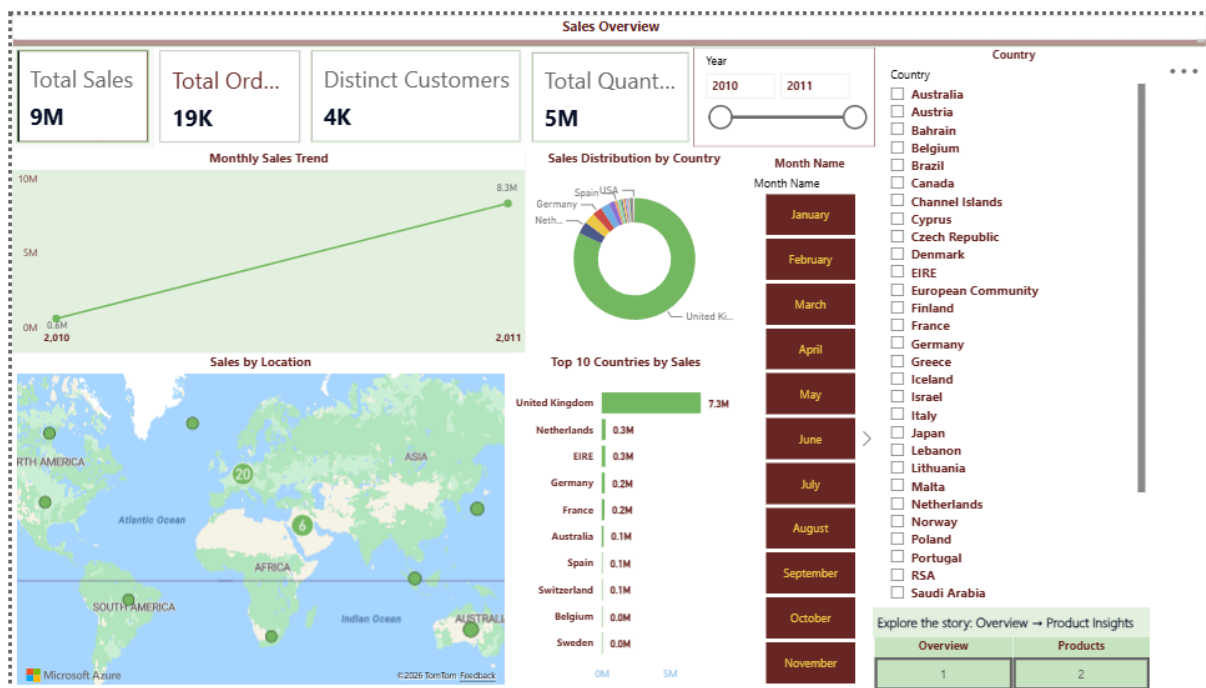
- Total Sales = SUM(FactSales[Sales])
- Total Orders = DISTINCTCOUNT(FactSales[InvoiceNo])
- Distinct Customers = DISTINCTCOUNT(FactSales[CustomerID])
- Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)
- % of Total Sales = DIVIDE(SUM(FactSales[Sales]), CALCULATE(SUM(FactSales[Sales]), ALL(FactSales)), 0)

7. Analysis and Visualizations (Power BI)

Report Pages

Page 1 – Sales Overview

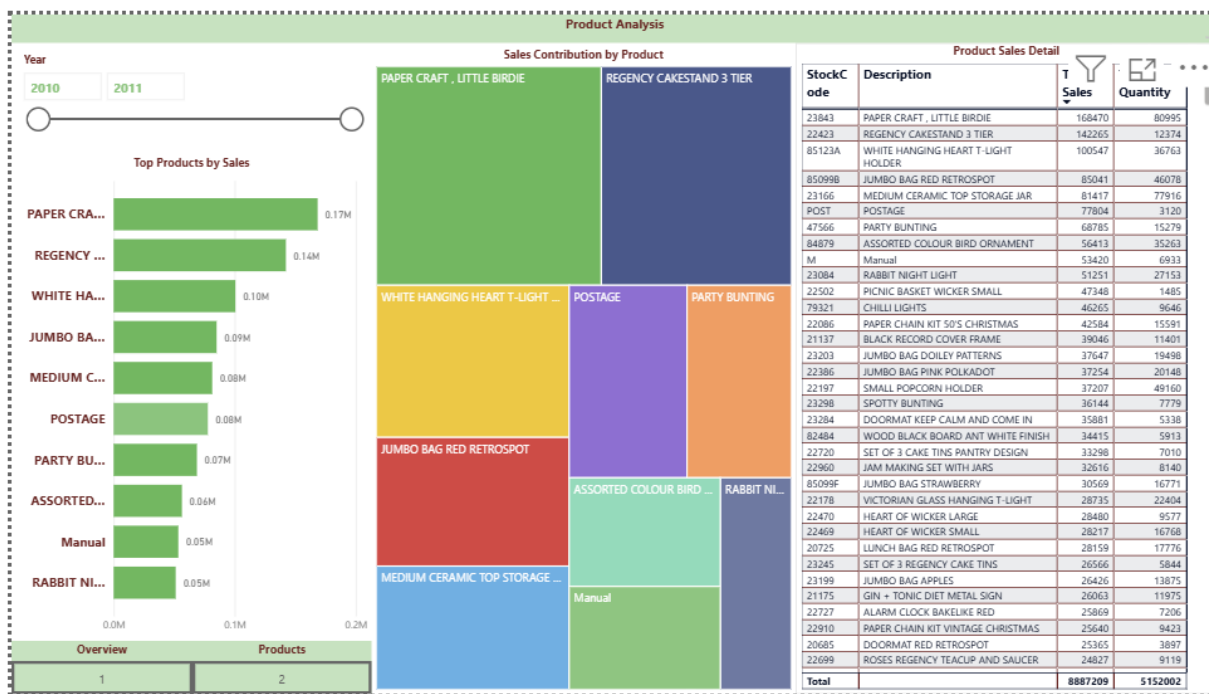
- KPI Cards: Total Sales, Total Orders, Distinct Customers, Total Quantity
- Line Chart: Monthly Sales Trend, with year-to-month drill down
- Donut Chart: Sales Distribution by Country
- Map: Sales by Location
- Bar Chart: Top 10 Countries by Sales
- Slicers: Year, Country, Month Name



Consolidated Dashboard – Sales Overview page

Page 2 – Product Analysis

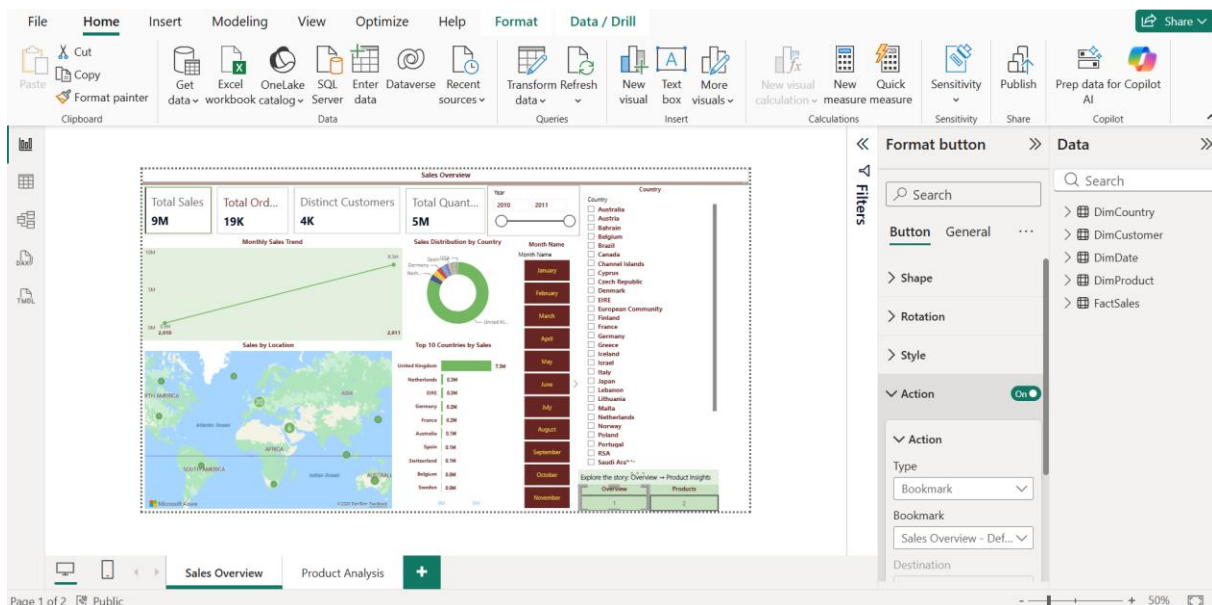
- Bar Chart: Top Products by Sales
- Treemap: Sales Contribution by Product
- Table: Product Sales Detail (StockCode, Description, Total Sales, Quantity)
- Top N filter: Top 10 products by Total Sales



Consolidated Dashboard – Product Analysis page

Interactive Storytelling Navigation (Bookmarks + Buttons)

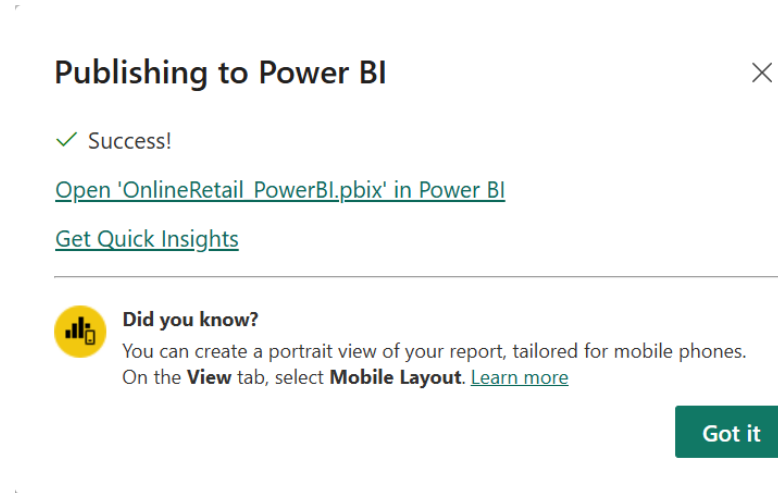
The Sales Overview and Product Analysis pages include a click-through navigation system built with Power BI Bookmarks and Buttons, so viewers can move between pages without using the page tabs.



Format button pane – button Action set to Type: Bookmark, target “Sales Overview - Default”

- Two bookmarks were created: “Sales Overview - Default” and “Product Analysis - Default”, each capturing the default filter/slicer state of its page.
- Two navigation buttons (“Overview” and “Products”) were added and linked to these bookmarks via Action > Type = Bookmark, then copied so both pages share identical navigation controls.

- A short narrative text box was added near the buttons: “Explore the story: Overview → Product Insights”.
- In Power BI Desktop, Ctrl+click triggers the bookmark while editing; after publishing or in Presentation Mode, a normal single click navigates as expected.



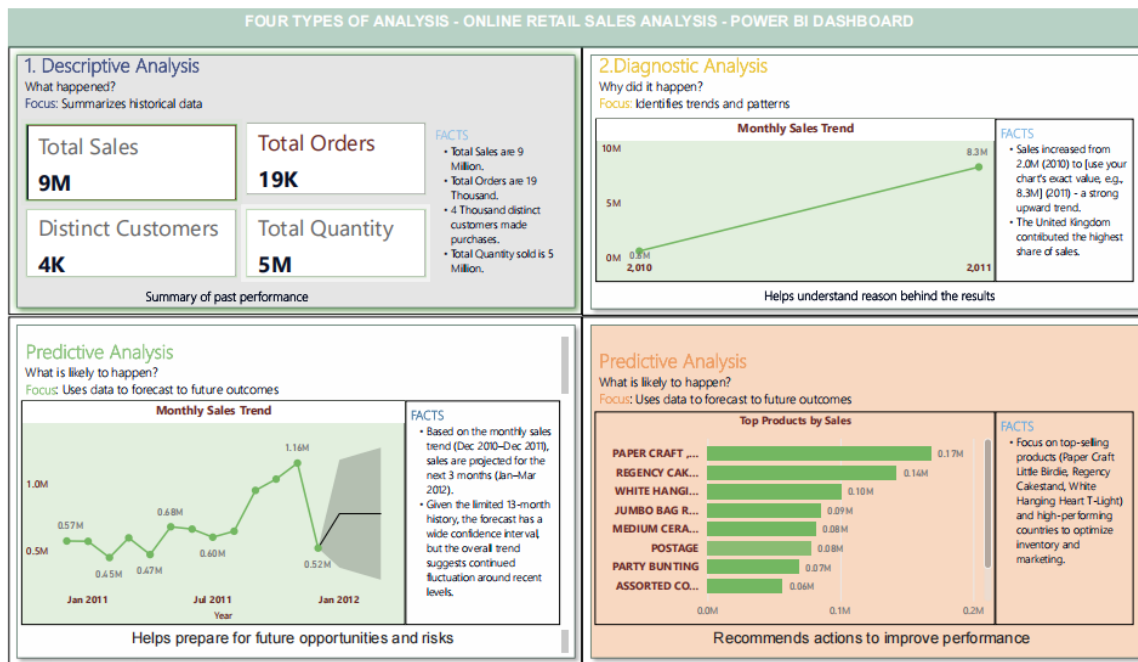
Publish success popup – report published to Power BI Service

- Published via File > Publish, then verified in Power BI Service that both buttons navigate correctly, with the browser URL updating with a bookmarkGuid parameter.
- Scheduled refresh was configured under Workspace > Dataset Settings, and key visuals were pinned to a dashboard for an at-a-glance summary view.

8. Insights & Conclusions

Key Findings

- Total sales for the period were about £9 million, from around 19,000 orders and 5 million units sold, across roughly 4,000 distinct customers.
- The United Kingdom made up about 82% of total sales (around £7.3 million), followed by the Netherlands, EIRE, Germany and France.
- “Paper Craft, Little Birdie” and “Regency Cakestand 3 Tier” were the top two selling products by revenue, at roughly £168K and £142K respectively.
- The average order value across the period was about £480.
- A small group of customers generated a disproportionate share of total sales.
- The sales trend shows a strong upward pattern from 2010 into 2011, with November 2011 as the peak month, consistent with Q4 holiday gifting.



Four types of analysis – Online Retail Sales Analysis, Power BI Dashboard

Analysis Report

Descriptive Analysis

- United Kingdom is the dominant market, contributing the large majority of total sales across the 37+ countries served.
- Total Sales reached 9 million, from 19 thousand orders, 4 thousand distinct customers and 5 million units sold.

Diagnostic Analysis

- Sales are heavily UK-concentrated because the business is a UK-headquartered retailer with secondary international wholesale demand.
- The best-selling products are mostly low-priced, high-volume items, which is why quantity sold is high relative to total sales value.
- A large share of the raw data quality issues (missing CustomerID, cancellations) originated from wholesale/bulk order entries and cancelled orders, which needed removal for accurate reporting.

Predictive Analysis

- Based on the monthly trend from December 2010 to December 2011, sales are projected to continue in a broadly similar range into the following months, with a wide confidence interval given the limited history.
- Top-performing products and countries (UK, Netherlands, EIRE, Germany, France) are likely to continue driving the majority of future revenue if current buying patterns persist.

Prescriptive Analysis

- Focus on retaining the top-spending customers, since a small group generates a large part of total revenue.
- Grow sales in next-best-performing countries such as the Netherlands, EIRE, Germany and France rather than spreading marketing resources thinly across all 37+ countries.

- Bundle popular, low-cost products such as “Paper Craft, Little Birdie” with slower-selling items to lift average order value.
- Verify whether the apparent drop at the end of the trend line reflects an actual decline or simply that the dataset ends partway through December 2011.

9. Conclusion

The Excel and Power BI integration proved effective for end-to-end data analysis, from a raw 541,909-row transaction file to a cleaned 392,692-row dataset and a fully working, published Power BI dashboard. Excel was used for data cleaning and transformation, and Power BI for data modelling, DAX measures and visualization.

Key tasks included establishing a star schema, creating calculated fields, building 12 DAX measures, and applying descriptive, diagnostic, predictive and prescriptive analysis across a three-page report with slicers, drill-down, bookmarks and click-through storytelling navigation — giving a clear view of sales, product and customer performance to support decisions on customer retention, market expansion and promotions.

Thank You!